

FORT SMITH, AR, USA

WMO: 723440

Lat: 35.333N

Lon: 94.363W

Elev: 449

StdP: 14.46

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13964

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.9	22.3	4.6	7.1	23.3	9.1	8.9	27.9	24.9	44.0	21.1	49.1	6.8	320	0.439

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	100.0	75.9	96.9	76.3	94.3	76.0	79.9	91.7	78.8	90.7	77.9	89.5	7.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.9	142.3	84.7	75.7	136.7	83.8	74.7	132.2	83.2	43.6	91.8	42.5	91.1	41.5	89.5	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		DB	WB
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.3	17.7	15.7	11.6	103.0	4.0	4.4	8.7	106.1	6.4	108.7	4.2	111.1	1.3	114.3		
(5)			10.0	81.5	3.7	1.6	7.4	82.6	5.2	83.5	3.2	84.4	0.5	85.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.6	40.6	44.9	53.5	62.4	70.6	78.9	82.9	82.4	75.0	63.8	52.2	42.9		
	DBStd	16.83	9.94	9.92	9.53	7.73	6.71	4.90	4.66	4.94	6.72	8.08	8.78	9.10		
	HDD50	936	321	196	72	7	1	0	0	0	0	4	77	258		
	HDD65	3063	757	563	372	135	28	0	0	0	9	122	391	686		
	CDD50	5540	29	55	180	381	639	867	1020	1006	751	431	143	38		
	CDD65	2193	1	1	15	58	202	417	555	541	310	84	8	1		
	CDH74	23773	2	17	169	590	1785	4417	6673	6287	2975	779	75	4		
	CDH80	11253	0	2	26	125	593	2055	3522	3352	1329	240	9	0		
(14) Wind	WSAvg	6.7	7.5	7.8	8.2	8.0	6.6	5.8	5.3	5.4	5.7	6.1	6.8	7.1		
(15) Precipitation	PrecAvg	45.9	2.4	2.7	4.2	4.5	5.8	4.1	3.4	3.1	4.1	4.0	3.9	3.5		
	PrecMax	78.5	6.5	7.1	12.3	9.6	19.8	9.3	10.4	9.5	15.9	12.4	13.7	11.3		
	PrecMin	27.1	0.2	0.6	0.5	0.6	1.1	0.4	0.0	0.0	0.0	0.0	0.3	0.4		
	PrecStd	10.9	1.5	1.6	2.4	2.3	3.3	2.2	2.8	2.2	3.0	2.9	2.9	2.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.8	76.3	83.3	87.2	92.3	98.8	103.9	104.4	99.5	90.0	80.9	73.4		
		MCWB	60.7	60.8	64.4	68.1	73.3	75.0	74.8	75.3	74.9	71.2	65.9	62.9		
	2%	DB	68.3	71.6	78.8	83.4	88.9	95.1	99.7	100.3	94.2	86.2	75.7	68.3		
		MCWB	58.4	59.5	62.6	66.1	72.5	76.1	76.3	76.0	74.3	69.6	63.0	59.3		
	5%	DB	62.6	67.2	74.6	80.6	86.2	92.5	96.7	97.2	91.3	82.3	72.1	62.9		
		MCWB	53.5	56.1	60.5	65.1	71.4	75.3	77.0	76.1	73.8	67.3	61.1	54.8		
	10%	DB	56.9	62.4	70.4	77.2	83.4	90.0	93.9	94.2	87.8	78.3	67.6	57.7		
		MCWB	49.3	52.6	58.1	63.5	70.4	74.8	76.9	75.8	72.2	65.6	58.0	51.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.0	64.8	67.4	71.9	77.1	80.2	81.5	81.2	78.7	75.0	68.8	64.8		
		MCDB	70.5	73.3	79.0	81.8	87.8	91.4	91.6	92.1	90.0	85.3	77.5	71.5		
	2%	WB	60.0	61.9	65.0	69.6	75.0	78.4	80.1	79.7	77.2	72.7	65.4	61.0		
		MCDB	66.4	69.5	76.1	79.1	84.8	90.0	91.8	91.9	88.8	81.6	72.8	67.7		
	5%	WB	55.2	58.1	62.6	68.0	73.4	77.3	79.0	78.5	75.8	70.7	62.9	56.6		
		MCDB	60.3	64.7	72.1	77.2	82.8	88.5	91.1	90.4	86.4	77.8	69.9	61.3		
	10%	WB	49.8	53.5	59.8	65.9	71.9	76.1	78.1	77.5	74.5	68.3	59.6	51.9		
		MCDB	56.3	60.2	67.9	74.6	80.8	86.6	89.6	89.1	84.0	75.4	66.0	56.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.7	20.5	21.7	22.3	19.6	18.9	19.6	20.6	21.2	22.3	21.4	18.7		
		MCDBR	28.8	28.8	28.1	26.5	22.7	21.4	23.2	24.9	24.4	26.1	26.7	26.3		
	5% WB	MCWBR	18.6	18.0	14.7	12.6	9.1	7.0	6.0	6.3	7.8	11.2	16.0	17.7		
		MCWBR	23.9	24.4	23.6	21.2	19.4	19.4	20.2	20.7	20.5	20.5	23.1	22.2		
(40) Clear-Sky Solar Irradiance	taub		0.315	0.319	0.356	0.392	0.425	0.443	0.464	0.462	0.403	0.355	0.329	0.306		
		taud	2.485	2.483	2.406	2.313	2.283	2.260	2.245	2.224	2.359	2.475	2.500	2.522		
	Ebn at Noon		282	294	290	283	274	268	261	259	271	277	274	278		
		Edh at Noon	27	30	35	41	43	43	44	44	36	30	26	24		
(44) All-Sky Solar Radiation	RadAvg		798	946	1292	1610	1773	2010	2006	1839	1567	1186	857	688		
	RadStd		76	129	82	95	162	211	124	121	116	160	101	89		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+241	+166
(47) Regional (28 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page